Week 5 · Part A — diagnoses of the nine charts

Written for: the marker of CDB601220 Week 5. Left panel = chart as the recipe says; right = redesign. All numbers are printed by partA_charts.py (partA_output.txt). Data is synthetic course data (fictional federation).

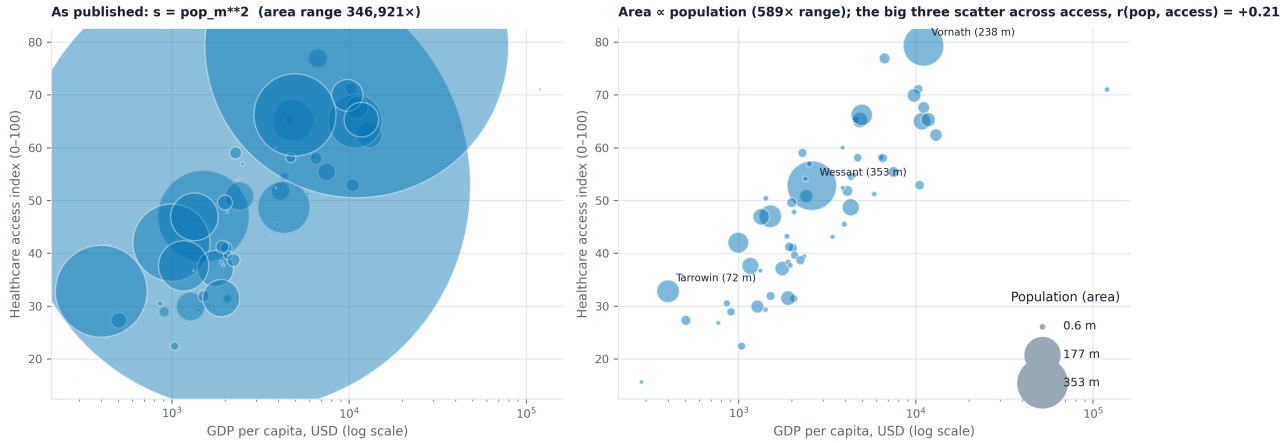
1 · “Population and healthcare access are unrelated”

Conclusion invited: Population size tells you nothing about access: the big marks are spread across the plot.

Decoy: 'the data really are unrelated'. The numbers are weakly related (r = +0.21, Spearman +0.22, p = 0.09), so the claim is not false, it is unreadable.

Mechanism (the size channel lies): scatter(s=) takes *area*, and the recipe feeds it pop_m**2, so area goes as population squared: the largest mark is 346,921x the smallest for a 589x population range (Lie Factor 589). Three countries fall under 1 pt², nine have under 2 m people. Note: the brief's self-check (589x) is what linear-in-population area gives; the literal recipe gives 346,921x, and I report both. The pattern the eye does see is GDP: r(access, log GDP) = +0.83.

Redesign: area linear in population via sizes=(12, 1500), a legend of real values, three labelled countries: the three biggest sit at access 52.8, 79.3 and 32.8.



Redesign: mark AREA linear in population, real-value size legend, three direct labels. Synthetic course data, CDB601220 Week 5 (fictional federation).

2 · “The outbreak is concentrated in the three districts around the capital”

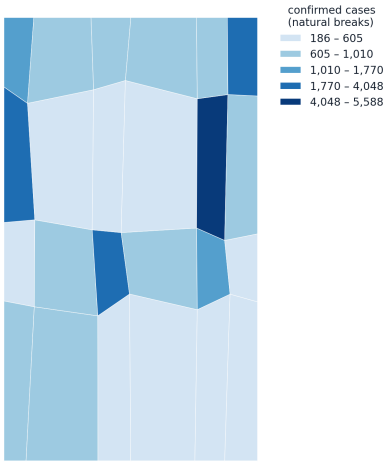
Conclusion invited: A cluster of disease around the capital: D17, D13, D09.

Decoy: 'counts are real counts'. True, and irrelevant: the question is risk.

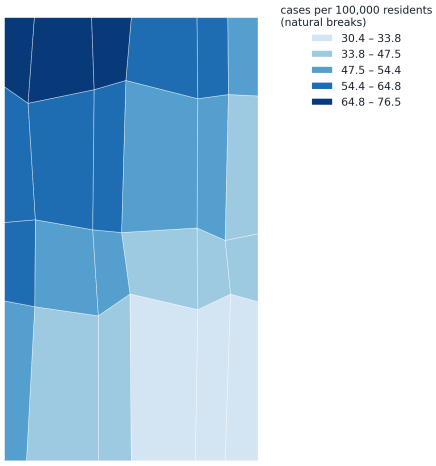
Mechanism (the denominator): the map shows cases, and cases follow people: r(cases, population) = +0.986. The three are not even adjacent (lon 72.6, 66.4, 69.4). Their rates are 51.0, 64.0 and 49.9 against a pooled 52.8 per 100,000: two are below average. By rate the top three are D19, D20, D21 (76, 74, 69): overlap with the count list = 0. 24 districts join with 0 failures.

Redesign: cases per 100,000 residents, equal-area projection, Jenks, 5 classes, breaks printed.

As published: cases (raw count)



Cases per 100,000 residents: the dark districts are D19–D21, not D17/D13/D09



Redesign: normalised by resident population; equal-area projection; Jenks 5 classes. Synthetic course data, CDB601220 Week 5 (fictional federation).

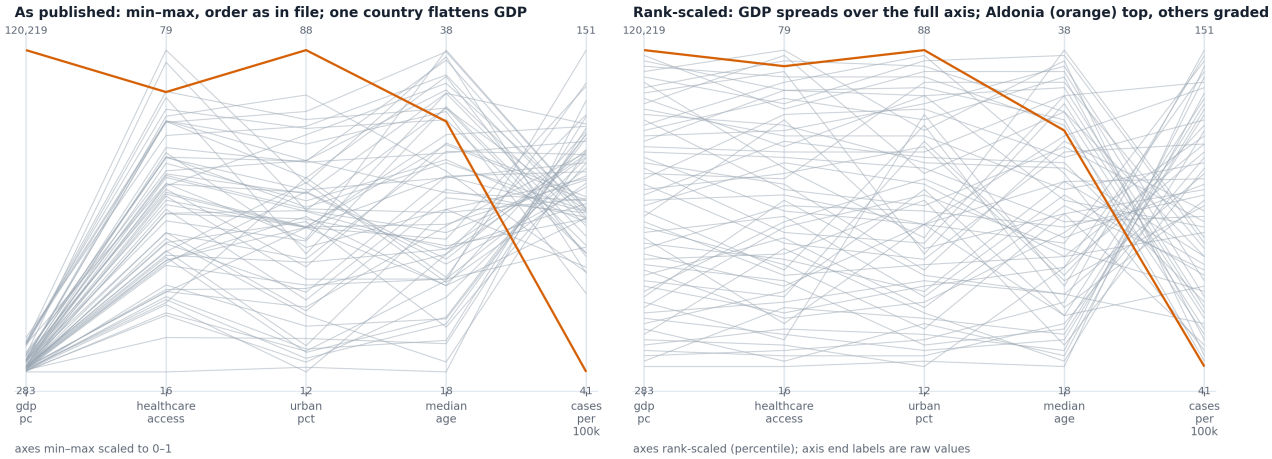
3 · “GDP per capita barely varies”

Conclusion invited: GDP is almost flat across countries.

Decoy: 'rich and poor countries are the same'. GDP runs $283–$13,009 (46x) once one country is set aside.

Mechanism (scaling): min–max divides by the range, and one country, Aldonia ($120,219), is 9.2x the second ($13,009). After scaling 58 of 60 sit below 0.10 and the median is 0.017. The axis is flat because the scaling is anchored to an outlier.

Redesign: percentile-rank scaling (median 0.51, only 5 of 60 below 0.10). Aldonia is brushed, not dropped, and the axis says 'rank scaled'.



Redesign: percentile-rank scaling (immune to the 9.2x outlier); outlier brushed, not dropped. Synthetic course data, CDB601220 Week 5 (fictional federation).

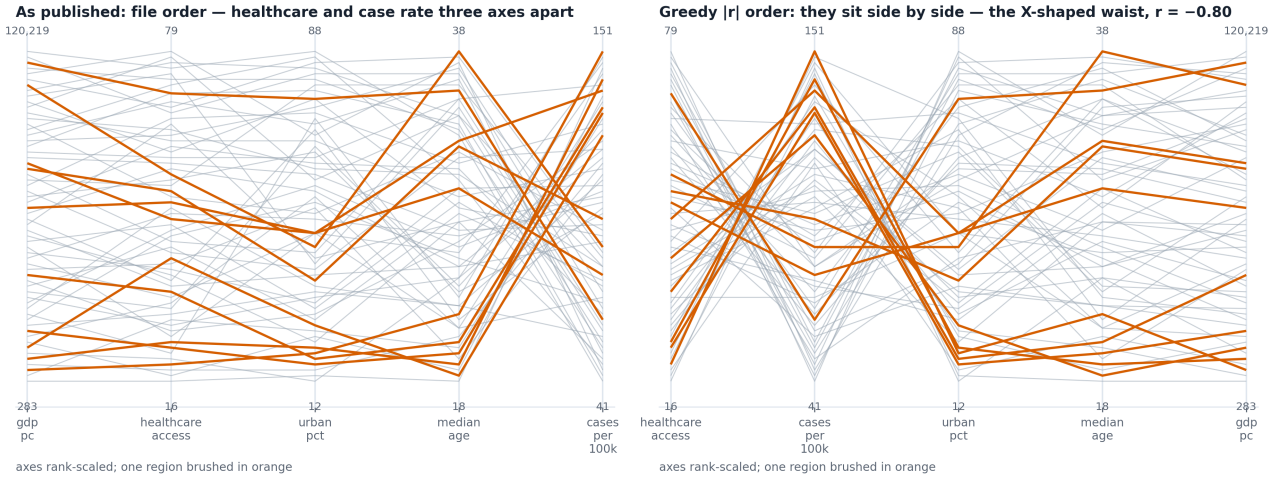
4 · “Healthcare access and case rate are unrelated”

Conclusion invited: No crossing pattern between the two.

Decoy: 'then they are unrelated'. They are the strongest pair of the ten: r = –0.799.

Mechanism (axis order): in file order the two axes are three apart (urban_pct and median_age between them), and a parallel coordinates plot only shows a relationship between *adjacent* axes. Order by the greedy |r| chain (healthcare_access → cases_per_100k → urban_pct → median_age → gdp_pc) puts them next to each other and the X-shaped waist appears.

Redesign: same plot, order from geolib.corr_chain, fixed before looking at the picture, rank-scaled.



Redesign: axis order from geolib.corr_chain, fixed before looking at the picture. Synthetic course data, CDB601220 Week 5 (fictional federation).

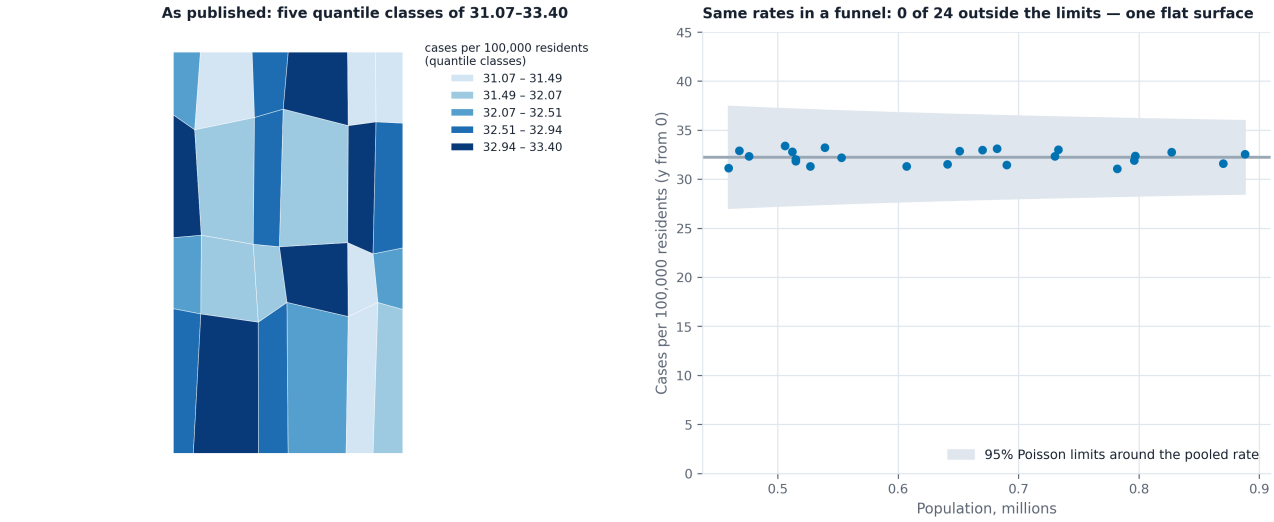
5 · “There is a clear five-level gradient in incidence”

Conclusion invited: Five tiers of risk across the federation.

Decoy: 'the colours show real differences'. The legend edges are only 0.4–0.6 per 100,000 apart.

Mechanism (classification): quantile classes force five equal-sized groups (5/5/4/5/5) whatever the data. The rate runs 31.07–33.40, a spread of 7.2% of the mean (CV 2.2%); 0 of 24 districts lie outside the 95% Poisson funnel. The 'gradient' is the classifier, not the disease. Not fixed by a different chart type: the decision to classify is the error.

Redesign: the same rates against the Poisson funnel, y axis from zero: one flat surface.



Redesign: the same rates tested against Poisson noise instead of forced into five classes. Synthetic course data, CDB601220 Week 5 (fictional federation).

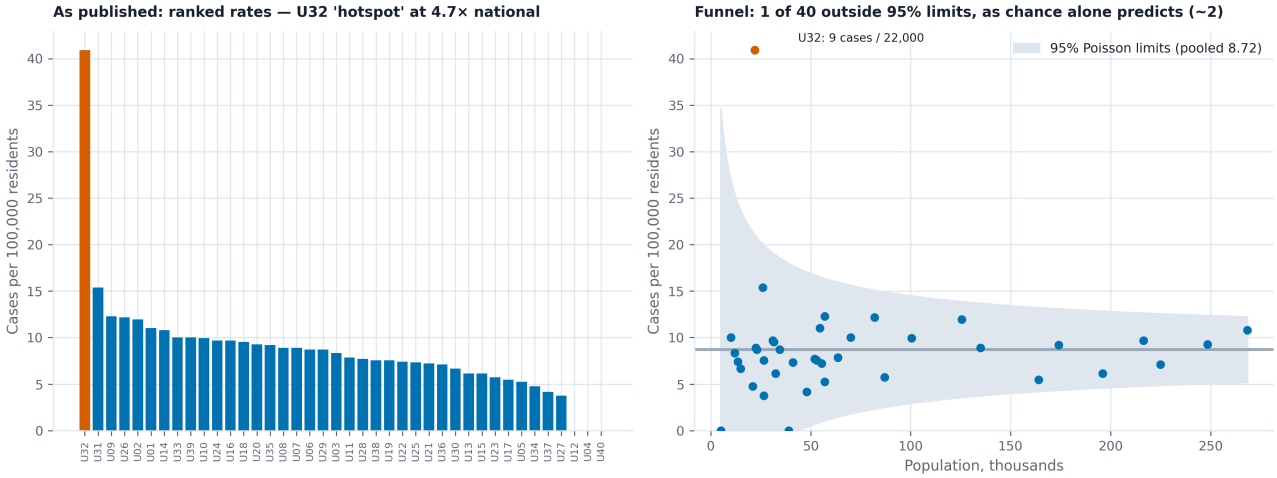
6 · “Unit U32 is a hotspot”

Conclusion invited: U32 runs at 4.7x the national rate.

Decoy: 'the rate is the rate'. It is 9 cases in 22,000 people.

Mechanism (small numbers): every unit has the same true rate. One case more or fewer moves U32's rate by 4.5 per 100,000 (11% of its own value). The 10 units over 100,000 people sit at 5.5–12.0. U32 is the one unit outside the 95% funnel; with 40 units about 2 are expected to be outside by chance. It is also above the 99.8% limit (upper 28.2 vs 40.9), so I do not call it nothing: it is a unit to watch, not a published hotspot.

Redesign: funnel plot of rate against population with Poisson limits.



Redesign: rate against the population behind it, with Poisson control limits (geolib.funnel_limits). Synthetic course data, CDB601220 Week 5 (fictional federation).

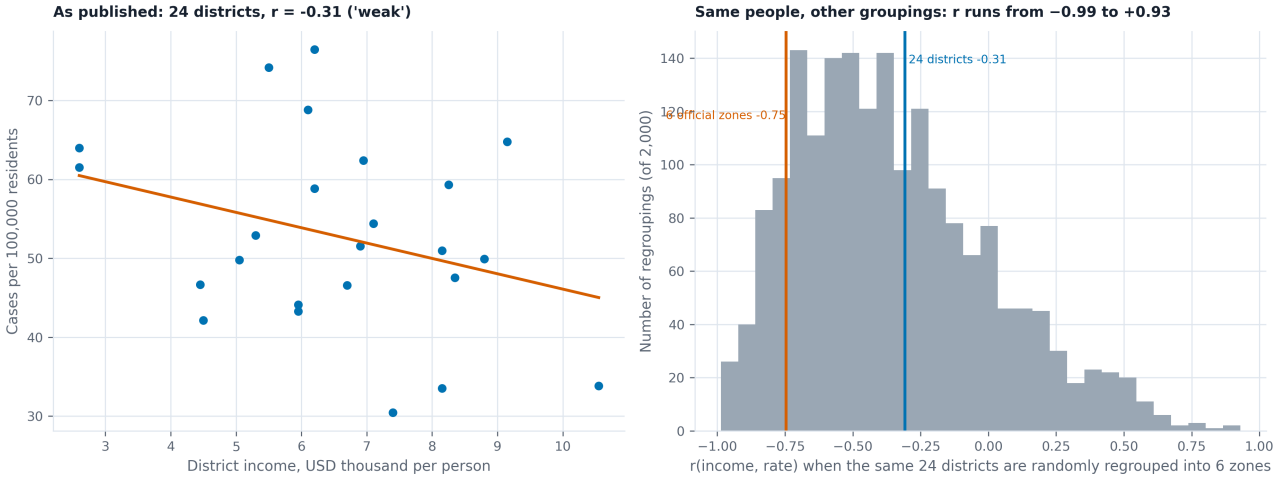
7 · “District income and case rate are only weakly related”

Conclusion invited: r = –0.31: a weak link.

Decoy: 'weak is weak'. Same people, different lines, a different answer.

Mechanism (the modifiable areal unit): merge the same 24 districts into the six official zones (population-weighted income) and r becomes –0.746. 2,000 random regroupings of the 24 into six zones give r from –0.99 to +0.93 (median –0.40). The coefficient belongs to the zoning. (districts_alt.geojson has no attribute table, so I did not score it: that would need areal interpolation, an assumption I will not make.) Not fixed by a different chart type: the decision is which units, and to report the range.

Redesign: the distribution of r across zonings, with 24-district and six-zone values marked.



Redesign: report the correlation's range across zonings, not a single number (MAUP, Openshaw 1984). Synthetic course data, CDB601220 Week 5 (fictional federation).

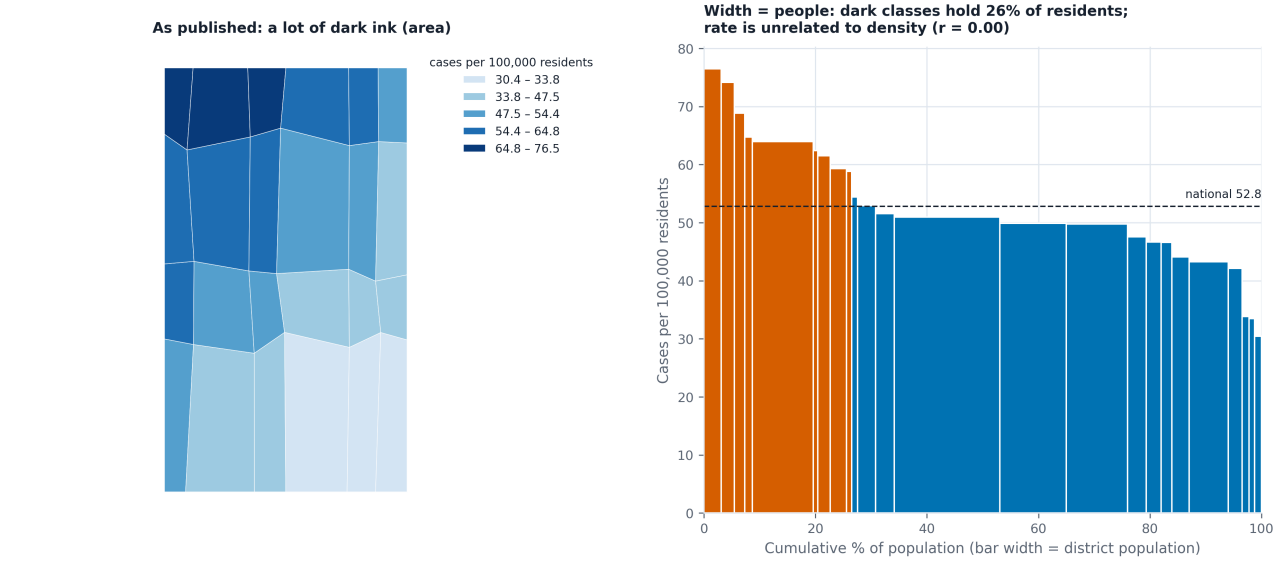
8 · “This is a rural outbreak — look how much of the map is dark”

Conclusion invited: Dark = rural.

Decoy: 'rural districts really have higher rates'. r(rate, density) = +0.001; denser half 53.0 vs sparser half 52.0 per 100,000.

Mechanism (ink follows area, not people): the five most populous districts are 12.6% of the land and 59.9% of the people. The top two classes cover 31.2% of the land but only 26.5% of residents (33.0% of cases). A choropleth gives each person in those five districts a twelfth of the ink.

Redesign: the same rates as bars with width proportional to population.



Redesign: the same rates with bar width ∝ population (ink goes to people, not land). Synthetic course data, CDB601220 Week 5 (fictional federation).

9 · “Region A is the stronger of the two”

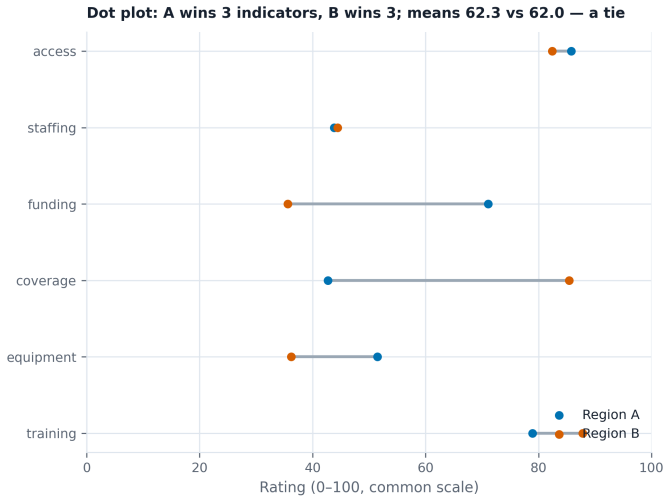
Conclusion invited: A encloses more area, so A is stronger.

Decoy: 'the polygon area is real'. It is real for one ordering.

Mechanism (angle and order): means are 62.3 and 62.0 (a 0.33 difference). File-order areas are 9,933 vs 9,433, but across all 60 distinct axis orders A is larger in 39 and B in 21; A's area ranges 9,366–10,443 and B's 8,564–10,555. The winner is decided by the axis order. A leads on access, funding and equipment; B on staffing, coverage and training.

Redesign: a dot plot on one common scale: three wins each, a tie.





Redesign: position on a common scale per indicator; no enclosed area. Synthetic course data, CDB601220 Week 5 (fictional federation).